

Chapter 43 — Exstrophy-Epispadias Complex — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Template** $\geq 3 \times 3$ cm → newborn closure; < 3 cm/polypoid/fibrotic → delay
- **Osteotomy** if closure > 72 h, diastasis > 4 cm or stiff pelvis
- **Gold-standard immobilisation**: external fixator + modified Buck traction
- **Staged repair**: closure birth → epispadias 6–12 mo → bladder neck 4–7 y
- **Bladder neck reconstruction** only at capacity ≥ 100 mL + motivated child
- **Best continence predictors**: template size + successful primary closure
- **Commonest closure error**: undivided urogenital diaphragm fibres
- **VUR (vesicoureteric reflux)** 100% after closure → continuous prophylaxis
- **Cloacal exstrophy**: osteotomy always; staged closure; spinal defect 85–100%
- **Not dry 3 h by 2 y** after bladder neck repair = failure

INCIDENCE AND INHERITANCE

- **Classic bladder exstrophy (CBE)**: 2.15/100,000 (USA); ≈ 1 in 46,000
- **Male:female** 2.3:1 (up to 5–6:1 in some series)
- **Family recurrence** ≈ 1 in 100; affected parent → **offspring 1 in 70** ($\approx 500\times$)
- **In vitro fertilisation (IVF)** $\approx 7.5\times$; early high-dose progesterone $\approx 10\times$ risk
- **Smoking** → **cloacal**; preconception folate → mildest (epispadias)
- **Commonest genetic variant**: 22q11.2 duplication
- **ISL1 (5q11.1)**: confirmed susceptibility gene
- **First cloacal exstrophy**: Schenck 1595; first epispadias: Heraclius

EMBRYOLOGY

- **Cause**: cloacal membrane not reinforced by mesoderm
- **Classic exstrophy** $> 50\%$ of the complex
- **Marshall-Muecke**: overdeveloped membrane, week 4; rupture timing sets variant

CLASSIC BLADDER EXSTROPHY

- **Posterior pelvis** externally rotated 12° ; **anterior** 18° , shortened 30%
- **Posterior segment length normal**; bone histology normal
- **Anterior corpora** $\approx 50\%$ short; posterior corpora normal length
- **Urethra anterior to prostate**; prostate never circumferential
- **PSA (prostate-specific antigen) below age norms** → still screen (digital rectal examination + PSA)
- **Vagina** ≤ 6 cm, normal calibre; cervix in anterior wall
- **Levator hiatus** $\approx 2\times$ wide; puborectalis angle $< 65^\circ$ in dry adults
- **Prolapse predictor**: adult diastasis width (osteotomy not protective)
- **Colorectal anomaly** $\approx 2\%$ (72 $\times$); commonest imperforate anus
- **Prolapse pre-closure** → close; **post-closure** → outlet obstruction → cystoscopy
- **Inguinal hernia**: common, bilateral → repair + explore contralateral at closure
- **Commonest renal anomaly**: duplex system
- **Commonest variant**: skin-covered (split symphysis)
- **Superior vesical fissure**: intact sphincter → continent, no bladder neck surgery
- **All variants except superior vesical fissure** → formal closure at birth
- **Cystitis glandularis** in polyps → adenocarcinoma → lifelong cytology + cystoscopy

PRENATAL DIAGNOSIS

- **Only $\approx 25\%$ of BEEC** (bladder exstrophy–epispadias complex) diagnosed prenatally
- **Five ultrasound signs**: no bladder filling, low umbilicus, wide rami, small genitalia, growing lower mass
- **Pubic diastasis** > 10 mm; cord-to-tubercle distance < 5 th centile

SURGICAL RECONSTRUCTION OF BLADDER EXSTROPHY

- **Newborn care**: cord tied 2-0 silk; plastic wrap; saline irrigation
- **Precludes closure**: penoscrotal duplication, hypoplastic bladder, bilateral hydronephrosis
- **Ectopic bowel in plate** = relative contraindication
- **Delayed closure timing**: ASAP, or 6–9 months
- **No growth by 6–12 mo** → excise; colon conduit or ureterosigmoidostomy
- **Mean diastasis 4.8 cm**; > 6 cm → staged pelvic closure
- **< 72 h + malleable** → no osteotomy; doubt → osteotomise
- **Commonest osteotomy**: combined anterior innominate + vertical iliac (supine)
- **Pubic stitch**: No. 2 nylon horizontal mattress (strongest)
- **Fixator 4–6 wk**; Buck traction ≈ 4 wk; **Bryant** only without osteotomy
- **Spica failure** $\approx 47\%$ vs fixator + traction $\approx 7.5\%$
- **Failure factors**: immobilisation < 4 wk; non-paediatric orthopaedic surgeon
- **Osteotomy complication**: transient left femoral nerve palsy
- **Epidural**: T10–T12; 0.25% bupivacaine 1 mL/kg + 1:200,000 adrenaline
- **Closure**: 2-cm plate strip; taper over 12 Fr; meatus 14 Fr
- **Malecot suprapubic 4 wk**; stents 10–14 d; **urethra unstented**
- **Combined closure + epispadias**: boys > 6 months only
- **Capacity < 30 mL at 1–2 y** → warn re continence surgery
- **Residual < 50 mL** + prophylaxis protects kidneys
- **Recurrent UTI (urinary tract infection) + full bladder** → cystoscopy for pubic stitch erosion
- **CPRE** (complete primary repair of exstrophy): $\approx 50\%$ reimplant in year 1; most hypospadiac
- **Erlangen**: total repair at 8 wk, no osteotomy
- **Mainz**: ureterosigmoidostomy at 2 y; sigma pouch
- **Kelly**: radical soft-tissue mobilisation; avoids osteotomy
- **Testosterone enanthate 2 mg/kg** intramuscularly, 5 + 2 wk pre-urethroplasty
- **Epispadias repair < 12 mo** → better capacity
- **Modified Cantwell-Ransley**: 18-mm strip; distal 1 cm left on glans; stent 8 Fr
- **Female closure**: osteotomy if > 4 cm; diastasis narrower than males

- **YDL** (Young-Dees-Leadbetter): mucosal strip 15 × 30 mm; suspend to rectus
- **Voiding trial**: clamp ≤1 h; fail → 8-Fr Foley 5 d
- **Growth after newborn closure**: ≈54 mL in 24 months

OUTCOMES AND RESULTS

- **Continence after staged repair**: ≈70% males, ≈74% females; YDL series 60–82%
- **Capacity >100 mL** → higher continence, dry in half the time
- **Dry by day ≈14 mo, night ≈23 mo**
- **Delayed closure without osteotomy** → ≈10% continent
- **Closure <1 y** → far less augmentation
- **High-volume centre**: >5 closures/year
- **Fistula site** after Cantwell-Ransley: penile base, not corona
- **Ischaemic penile loss** → stop, immediate osteotomy

FAILURES AND COMPLICATIONS

- **Reclosure wait ≥4 months** (4–6)
- **Voided continence after reclosure** ≈1 in 6
- **Stricture**: within 60 d; watch for unexplained rectal prolapse
- **Diversion >6 mo** → augmentation; <6 mo → usually bowel-free
- **Vesicocutaneous fistula** = failed closure until proved otherwise
- **Redo YDL** only if patulous neck, >100 mL, stable urodynamics
- **Bulking** helps only if already dry 1–3 h
- **Transection + augmentation**: bladder stones ≈30%
- **Ureterosigmoidostomy**: needs normal anal continence; yearly colonoscopy
- **Too small to close** → non-refluxing colon conduit

ADOLESCENT AND ADULT CONCERNS

- **Augmentation cancer surveillance** from 5 years; smoking adds risk
- **Male paternity** ≈24%, half via ART (assisted reproductive technology)
- **Infertility cause**: verumontanum injury, retrograde ejaculation
- **Uterine prolapse**: mean age 16 y → sacral suspension, both sides
- **Delivery**: planned caesarean, urologist present
- **Continence after age 5** → more acting-out behaviour

EPISPADIAS

- **Male 1 in 117,000; female 1 in 484,000**
- **≈70% male** complete (penopubic)
- **Incontinence**: penopubic ≈all, penile most, glanular none
- **VUR** male 30–40%; female 30–75%
- **Treat as exstrophy** if diastasis >3 cm → osteotomy
- **Bladder neck** at ≈100 mL, age 4–5 (YDL + Marshall-Marchetti-Krantz + reimplant)
- **Continence after bladder neck reconstruction**: males 47–82%; females 70–87%
- **Davis (female)**: vesicular → subsymphyseal → retrosymphyseal

- **First cure** of male incontinence: Young 1922

CLOACAL EXSTROPHY

- **Incidence** 1 in 200,000–400,000; sex ≈1:1
- **OEIS**: omphalocele, exstrophy, imperforate anus, spinal defects
- **Spinal defect 85–100%**; lumbar ≈80%; lipomeningocele commonest
- **Spinal ultrasound** ≈ MRI (magnetic resonance imaging) in infants
- **Omphalocele 88–100%** → neonatal closure
- **Short gut** ≈25% even with normal length → keep hindgut
- **Renal anomaly 41–60%**; solitary kidney commonest (30× CBE)
- **Commonest müllerian anomaly**: uterine duplication
- **Prenatal diagnosis** ≈15%; Austin major: absent bladder commonest
- **Elephant-trunk sign**: prolapsed terminal ileum
- **Manzoni grid**: coded anatomy classification
- **Gender**: assign congruent with karyotype
- **Osteotomy success**: combined 90% > posterior 76% > anterior 61%
- **Staged closure** ≈4× odds success; single-stage fails ≈48% vs 15%
- **Primary failure 31–43%**; spica failure ≈72%
- **Diastasis >10 cm** → osteotomy + gradual fixator closure
- **Buck traction + fixator 6 wk** = gold standard
- **Median 2 continence procedures**; continent ≈11 y
- **Disability determinant**: neurological level; tethering in ≤33%
- **Ricketts score**: 6-point bowel + bladder continence

CLASSIC TRAPS

- **Trap**: penis short only from diastasis — **Actually**: anterior corpora 50% deficient
- **Trap**: lengthening adds corporal tissue — **Actually**: apparent length only
- **Trap**: all variants do better — **Actually**: only superior vesical fissure
- **Trap**: osteotomy prevents uterine prolapse — **Actually**: diastasis width predicts
- **Trap**: CPRE avoids bladder neck surgery — **Actually**: most need outlet procedure
- **Trap**: osteotomy alone secures closure — **Actually**: spica still fails ≈30%
- **Trap**: bladder neck repair early — **Actually**: wait for 100 mL and readiness
- **Trap**: epispadias continence from pubertal prostate — **Actually**: pelvic floor maturation
- **Trap**: exstrophy bladder neuropathic — **Actually**: immature, normal potential
- **Trap**: osteotomy improves cloacal continence — **Actually**: closure only, not continence